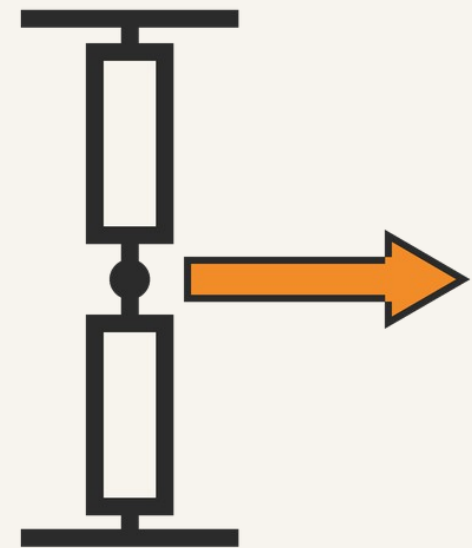


BOOKLET 3 • ELECTRONICS AND ANALOGUE
CONTROL

Voltage Dividers and Sensors

Two resistors in series, and the voltage
you take from between them.

National 5 Engineering Science



Daily Review

Lesson 4 — Input and Output Devices

- 1 State the rule for total resistance in series.
- 2 What does an LDR react to?
- 3 Describe how a thermistor's resistance changes as it gets hotter.
- 4 State Ohm's Law.
- 5 Name an output transducer that produces sound.

PART 1

The Voltage Divider

A supply gives one voltage. A divider gives you any smaller one you want.

By the end of this part...

- 1 I can describe how a voltage divider works
- 2 I can calculate a voltage in a fixed divider

Analogue signals

Booklet p.21

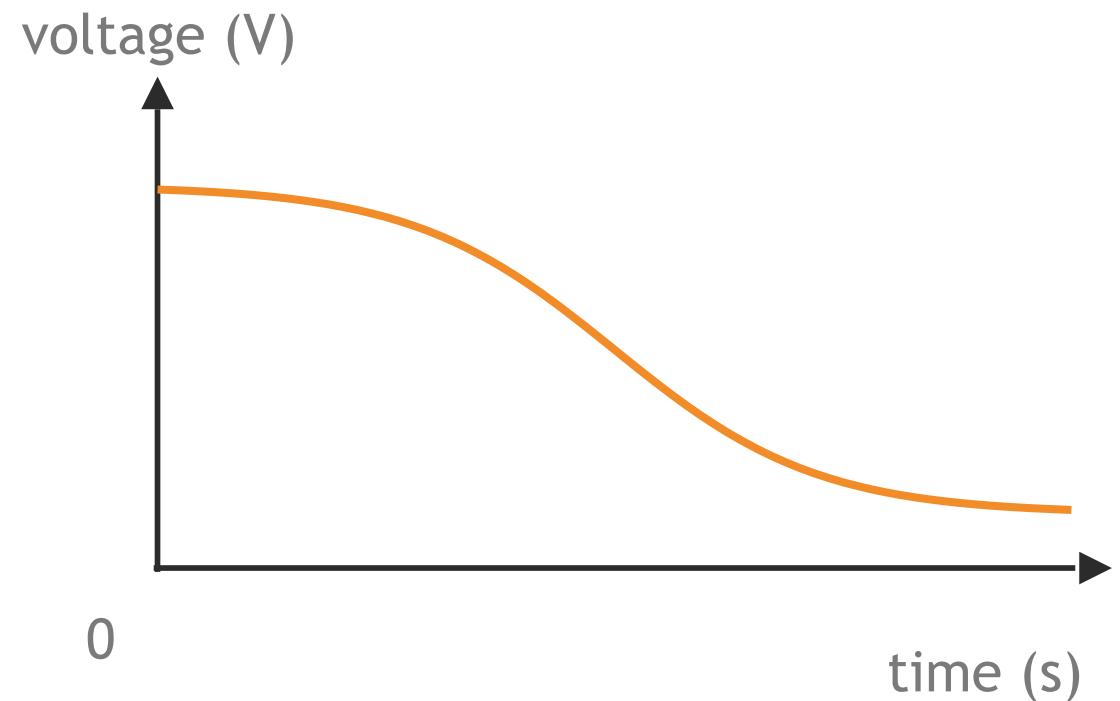
Analogue

A signal that can take any value in a range, and changes smoothly.

The example

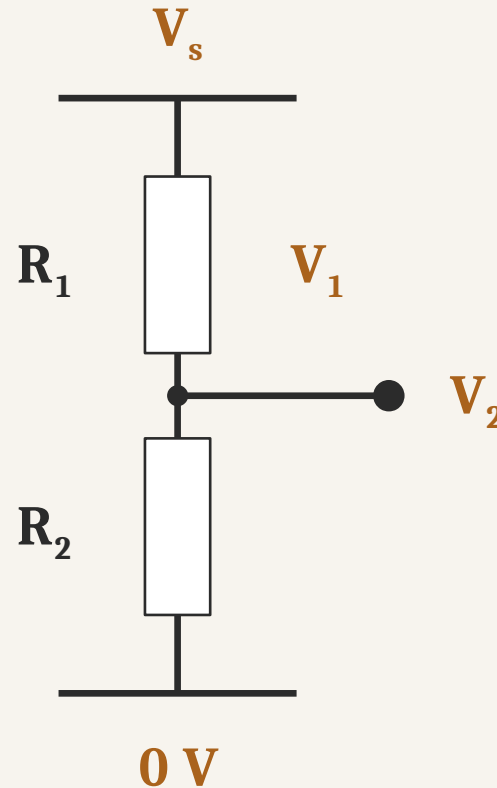
The voltage across an LDR divider as the light drops at sunset.

At sunset: a smooth fall



What a voltage divider is

Booklet p.21 — two resistors in series across the supply



The output V_2 is taken across the LOWER resistor, R_2 .

The bigger resistance takes the bigger share

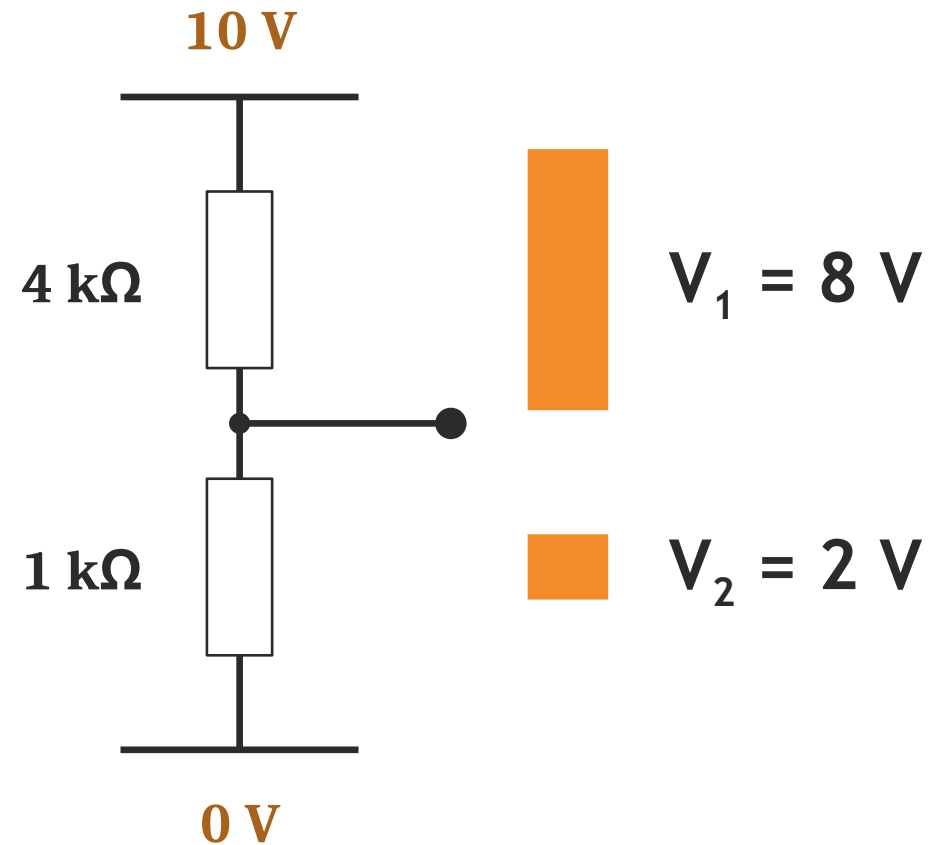
Booklet p.21

The rule

The supply is shared between the two resistors, in the same ratio as their resistances.

So

V_2 is always less than the supply voltage — some of it is always across R_1 .



The voltage divider relationship

Booklet p.21 – on the data sheet

$$V_1/V_2 = R_1/R_2$$

V_1, V_2

the two voltages, in volts (V)

R_1, R_2

the two resistances, in ohms
(Ω)

Worked example – find the output · p.21

In a voltage divider $R_1 = 3 \text{ k}\Omega$ and $R_2 = 1 \text{ k}\Omega$.
The voltage across R_1 is 9 V. Calculate V_2 .

$$V_1 = 9 \text{ V}$$

$$R_1 = 3000 \text{ }\Omega$$

$$R_2 = 1000 \text{ }\Omega$$

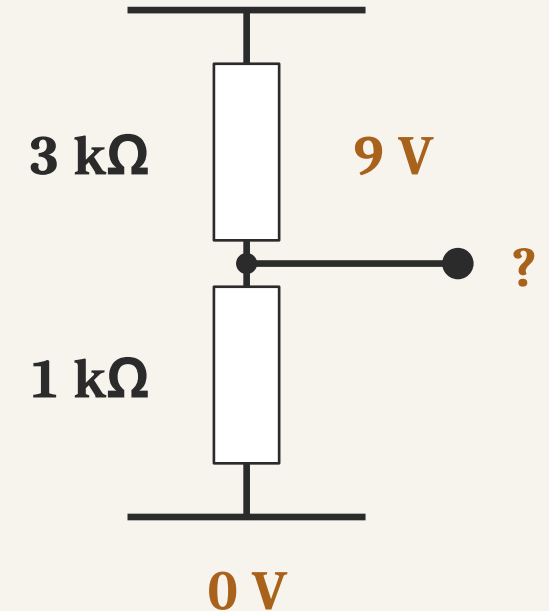
$$V_2 = ?$$

$$V_1/V_2 = R_1/R_2$$

$$V_2 = (V_1 \times R_2)/R_1$$

$$V_2 = 9 \times 1000/3000$$

$$\underline{V_2 = 3 \text{ V}}$$



Worked example – work backwards · p.21

In a voltage divider $R_1 = 5 \text{ k}\Omega$ and $R_2 = 2 \text{ k}\Omega$.
The voltage across R_2 is 2 V. Calculate V_1 .

$$V_2 = 2 \text{ V}$$

$$R_1 = 5000 \text{ }\Omega$$

$$R_2 = 2000 \text{ }\Omega$$

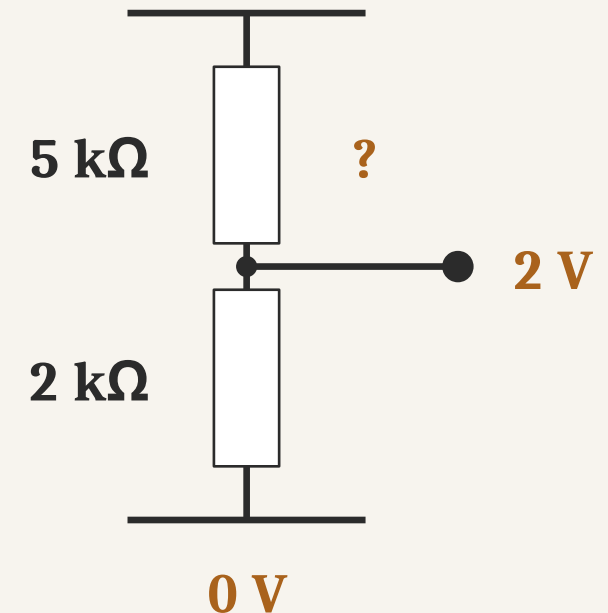
$$V_1 = ?$$

$$V_1/V_2 = R_1/R_2$$

$$V_1 = (V_2 \times R_1)/R_2$$

$$V_1 = 2 \times 5000/2000$$

$$\underline{V_1 = 5 \text{ V}}$$



Show me – divider calculations

SHOW ME BOARDS

1 $R_1 = 1 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $V_1 = 4 \text{ V}$. Find V_2 .

2 $R_1 = 2 \text{ k}\Omega$, $R_2 = 4 \text{ k}\Omega$, $V_1 = 3 \text{ V}$. Find V_2 .

3 $R_1 = 6 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $V_1 = 6 \text{ V}$. Find V_2 .

4 $R_1 = 1 \text{ k}\Omega$, $R_2 = 3 \text{ k}\Omega$, $V_2 = 9 \text{ V}$. Find V_1 .

5 The two resistors are equal. What fraction of the supply is V_2 ?

Show me – divider calculations

SHOW ME BOARDS

EXTRA PRACTICE

1 $R_1 = 500\ \Omega$, $R_2 = 500\ \Omega$, $V_1 = 5\ \text{V}$. Find V_2 .

2 $R_1 = 3\ \text{k}\Omega$, $R_2 = 6\ \text{k}\Omega$, $V_1 = 2\ \text{V}$. Find V_2 .

3 $R_1 = 8\ \text{k}\Omega$, $R_2 = 2\ \text{k}\Omega$, $V_1 = 8\ \text{V}$. Find V_2 .

4 $R_1 = 2\ \text{k}\Omega$, $R_2 = 6\ \text{k}\Omega$, $V_2 = 12\ \text{V}$. Find V_1 .

5 R_2 is three times R_1 . What fraction of the supply is V_2 ?

Your turn – TRY THIS

Booklet p.21 – TRY THIS: Voltage Divider

Answer the three questions in your booklet.
3 marks each: relationship, substitution,
then the answer with its unit.
Question 3 asks for V_1 , not V_2 .

Booklet p.21 – TRY THIS: Voltage Divider

TRY THIS — Answers

ANSWERS

Booklet p.21 — Voltage Divider

Q	Working	Answer
1	$V_2 = (6 \times 1000) \div 2000$	3 V
2	$V_2 = (4 \times 940) \div 470$	8 V
3	$V_1 = (4 \times 1000) \div 2000$	2 V

In question 2, R_2 is the bigger resistor — so V_2 is the bigger voltage.

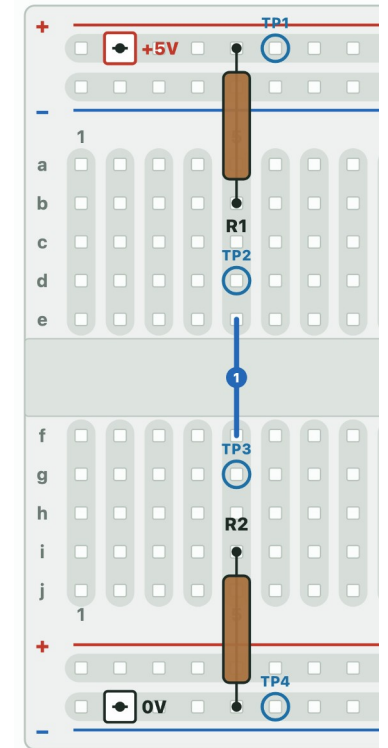
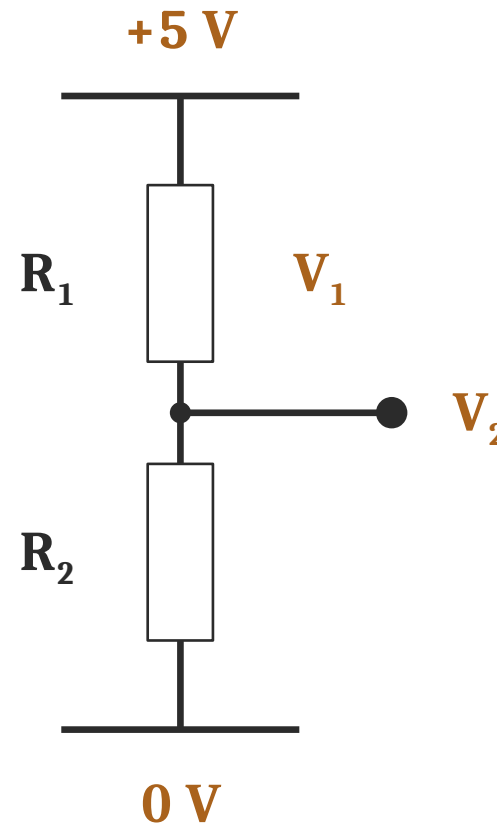
Checkpoint

- 1 What is a voltage divider made from?
- 2 Which resistor is the output voltage taken across?
- 3 Two equal resistors divide a 10 V supply. What is V_2 , and why?

Practical 1 – The Fixed Voltage Divider

Practical sheet on Teams · success criteria 4.1, 4.2 and 2.6

Four pairs of resistors on a 5 V supply.
Predict the ratio, simulate it in Yenka, then build it.
 R_1 , the jumper and R_2 run down column 5 of the board.



pair A on the board

Daily Review

Part 1 — The Voltage Divider

- 1 What is the output of a voltage divider taken across?
- 2 $R_1 = 2 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $V_1 = 3 \text{ V}$. Find V_2 .
- 3 Can V_2 ever be bigger than the supply voltage?
- 4 State the voltage divider relationship.
- 5 Which resistor has the bigger voltage across it?

PART 2

Sensors in the Divider

Replace one resistor with an LDR and the output voltage follows the light.

By the end of this part...

- 1 I can use a variable voltage divider to provide an input signal
- 2 I can link a change in sensor resistance to a change in output voltage

Swap a resistor for a sensor

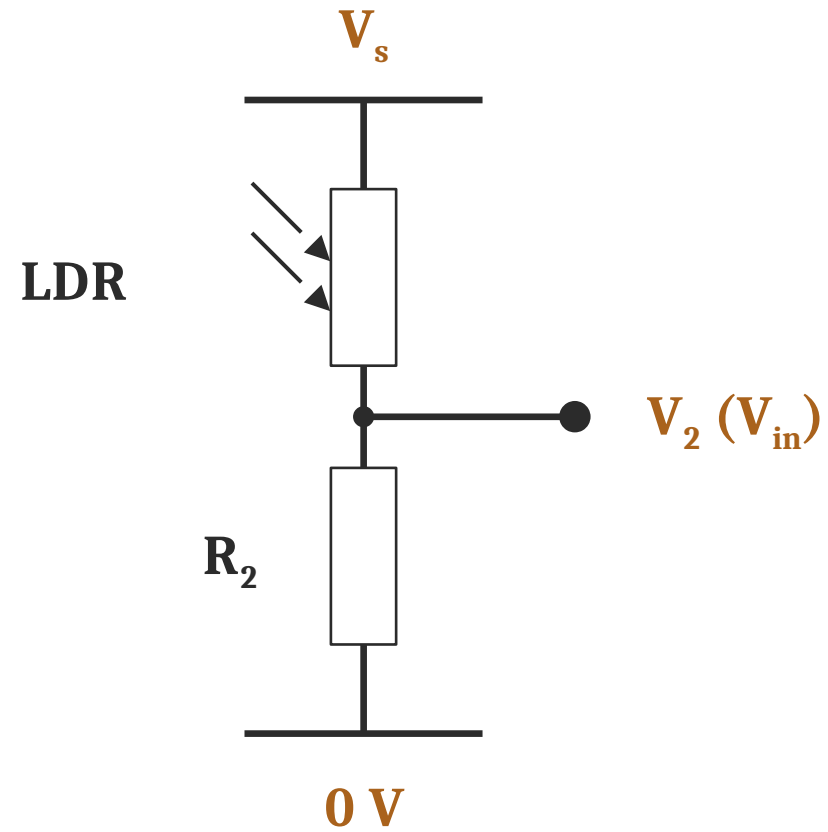
Booklet p.23 – the LDR takes the place of R_1

Light dependent resistor

The LDR. In the exam, write its full name.

As its resistance changes

so does V_2 . Exam papers call this voltage V_{in} .



LDR in the top position

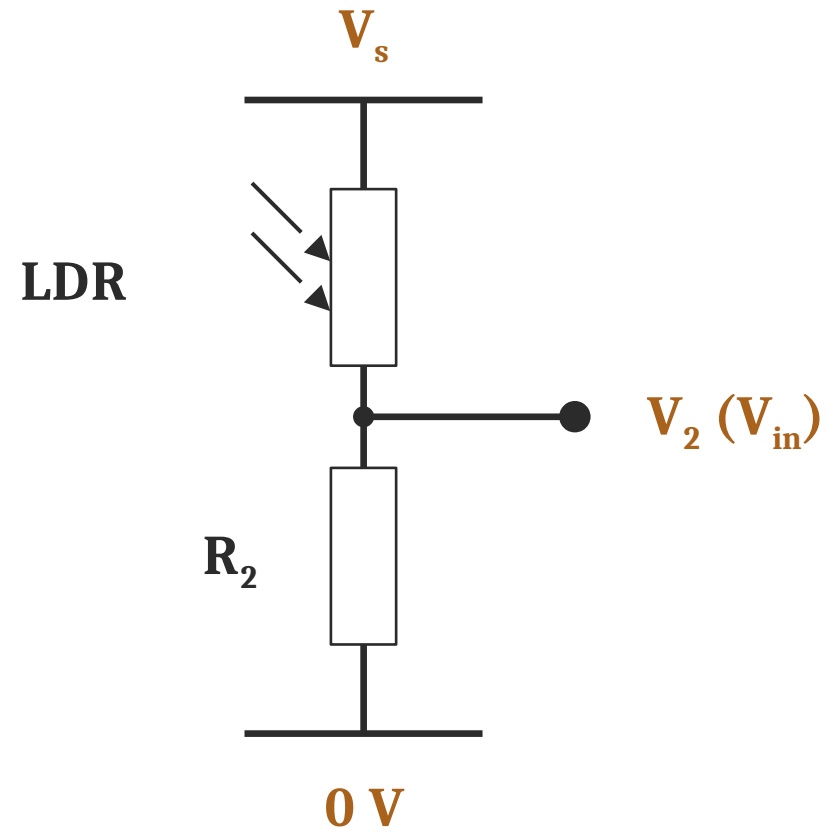
Booklet p.23 – the LDR is R_1

In bright light

The LDR's resistance is LOW,
so it takes little of the supply
– V_2 is HIGH.

In darkness

The LDR's resistance is HIGH,
so it takes most of the supply
– V_2 is LOW.



LDR in the bottom position

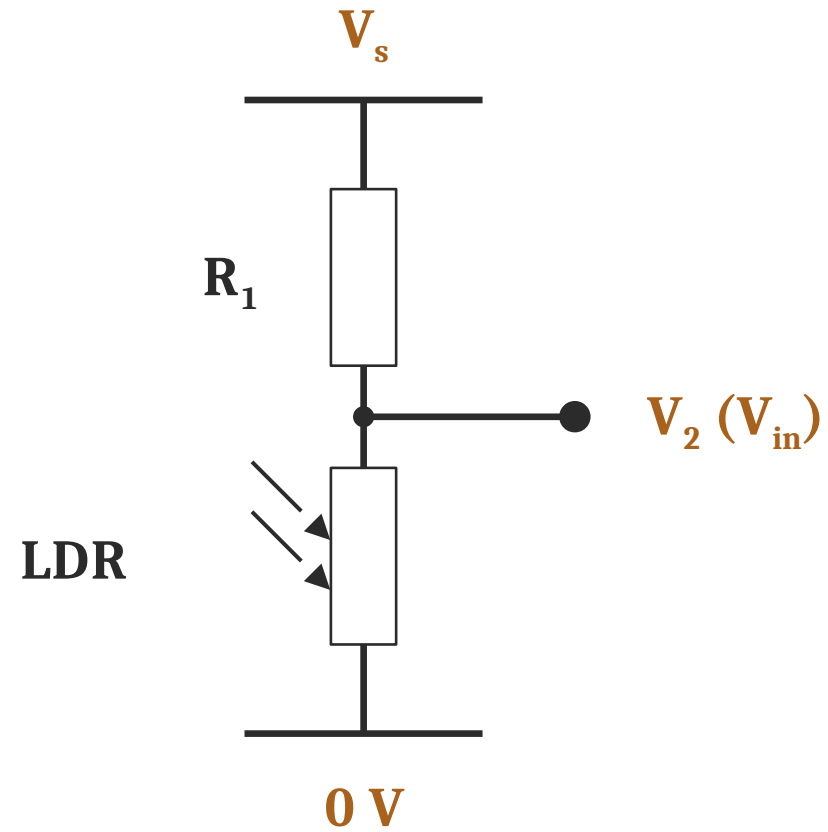
Booklet p.23 – the LDR is R_2

In bright light

The LDR's resistance is LOW, so little of the supply is across it – V_2 is LOW.

In darkness

The LDR's resistance is HIGH, so most of the supply is across it – V_2 is HIGH.



Where you meet it

Sensors in a divider, doing a real job

Street lights

A light sensor in a divider switches the lamp on as it gets dark.

Greenhouses

A thermistor in a divider opens the vents as it gets hot.



Photo: Bidgee,
CC BY 3.0, Wikimedia Commons



Photo: Kattegattt,
CC BY-SA 4.0, Wikimedia Commons

How to describe a sensor circuit

Booklet p.31 – the routine

First

what happens to the SENSOR'S RESISTANCE.

Then

what happens to V_2 (V_{in}).

Then

what the output device does.

LDR at the top, light rising

the light level rises



the LDR's resistance falls



V_2 (V_{in}) rises



the output switches on

Watch out

THE MISTAKE

- It gets darker, so V_2 (V_{in}) rises.
- Scores 1 mark of 2 in the exam.

THE CORRECT ANSWER

- The LDR's resistance rises, so V_2 (V_{in}) rises.
- Resistance first – every time.

Worked example – an LDR divider • p.23

In normal light an LDR (R_1) is **8 k Ω** and R_2 is 2 k Ω .
The voltage across the LDR is 8 V. Calculate **V_2** .

$$V_1 = 8 \text{ V}$$

$$R_1 = 8000 \text{ } \Omega$$

$$R_2 = 2000 \text{ } \Omega$$

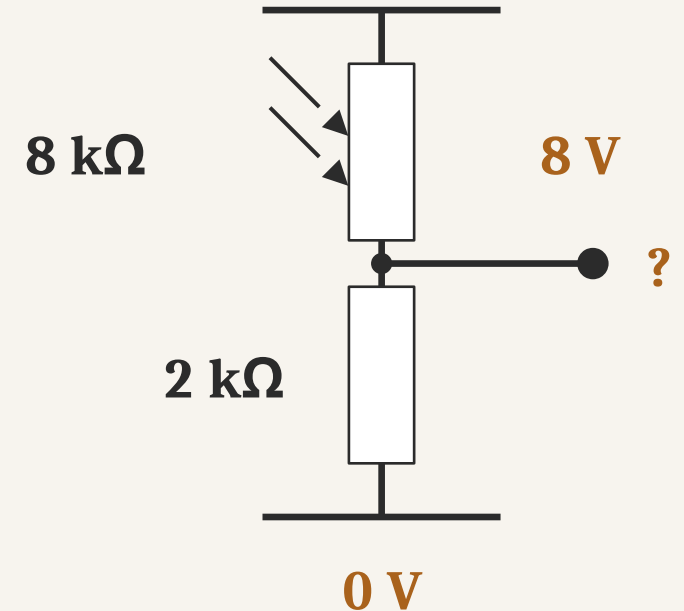
$$V_2 = ?$$

$$V_1/V_2 = R_1/R_2$$

$$V_2 = (V_1 \times R_2)/R_1$$

$$V_2 = 8 \times 2000/8000$$

$$\underline{V_2 = 2 \text{ V}}$$



Show me – which way does V_2 go?

SHOW ME BOARDS

- 1 The LDR is R_1 . It gets darker. What happens to V_2 ?
- 2 The LDR is R_2 . It gets darker. What happens to V_2 ?
- 3 A thermistor is R_2 . It gets hotter. What happens to V_2 ?
- 4 A thermistor is R_1 . It gets colder. What happens to V_2 ?
- 5 You want V_2 to rise as a room warms. Where does the thermistor go?

Show me – which way does V_2 go?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 State the full name of the component called an LDR.
- 2 The LDR is R_2 . It gets brighter. What happens to V_2 ?
- 3 A thermistor is R_2 . It gets colder. What happens to V_2 ?
- 4 A thermistor is R_1 . It gets hotter. What happens to V_2 ?
- 5 You want V_2 to rise as it gets dark. Where does the LDR go?

Your turn – TRY THIS

Booklet p.23 – TRY THIS: Sensor Voltage Divider

Answer the three questions in your booklet.
Read carefully which component is R_1
and which is R_2 – it changes the answer.

Booklet p.23 – TRY THIS: Sensor Divider

TRY THIS — Answers

ANSWERS

Booklet p.23 — Sensor Voltage Divider

Q	Working	Answer
1	$V_2 = (4 \times 3000) \div 2000$	6 V
2	$V_2 = (2 \times 20\,000) \div 10\,000$	4 V
3	$V_2 = (3 \times 6000) \div 2000$	9 V

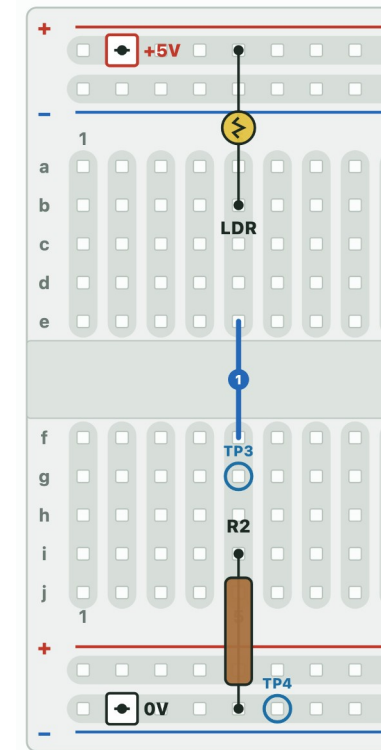
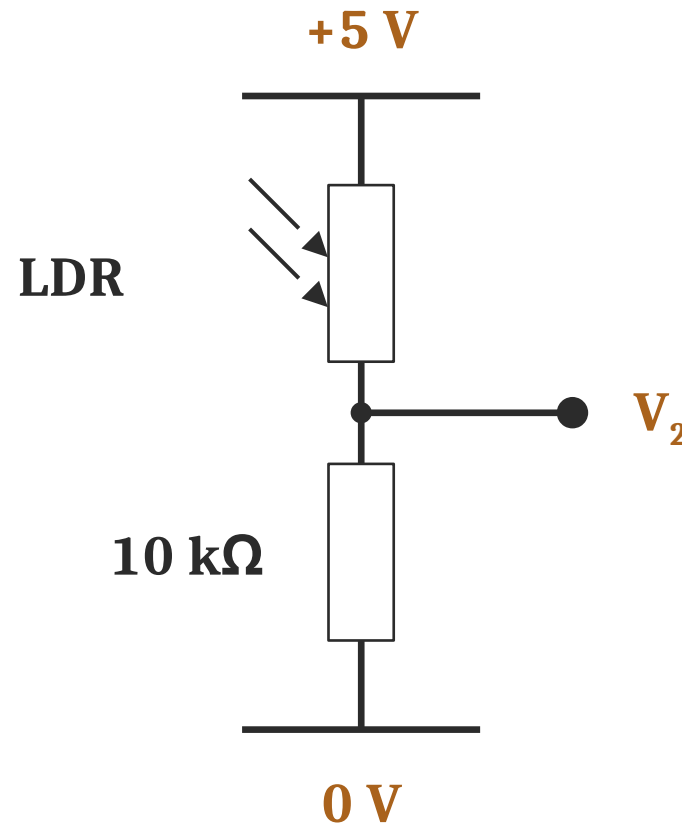
Every one has the bigger resistance taking the bigger voltage.

Practical 2 – The Sensor Voltage Divider

Practical sheet on Teams · success criteria 4.3, 4.4 and 4.5

An LDR and a thermistor, each at the top and at the bottom.

Predict, simulate in Yenka, then build and measure V_2 (V_{in}).



LDR at the top

Daily Review

Part 2 — Sensors in the Divider

- 1 The LDR is R_2 and it gets darker. What happens to V_2 ?
- 2 Which resistor is V_2 taken across?
- 3 $R_1 = 4 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $V_1 = 8 \text{ V}$. Find V_2 .
- 4 State the three steps for describing a sensor circuit.
- 5 What is an analogue signal?

PART 3

Reading a Sensor Circuit

The exam gives you a circuit and a data table. Both have to be read the same way.

By the end of this part...

- 1 I can interpret data tables for an LDR or thermistor
- 2 I can calculate an unknown resistance in a voltage divider

Reading a data table

Booklet p.23

Step 1

Find the sensor's resistance at the condition the question names.

Step 2

Put that resistance into the divider relationship with the fixed resistor.

Light level	Resistance
very dark	100 k Ω
dim	8 k Ω
room light	2 k Ω
bright sun	200 Ω

Dim — so use 8 k Ω .

Worked example – the missing voltage first

A divider on a 12 V supply has $R_1 = 2.2 \text{ k}\Omega$ and an LDR as R_2 .
 $V_2 = 9.0 \text{ V}$. Calculate the resistance of the LDR.

$$V_s = 12 \text{ V}$$

$$V_2 = 9.0 \text{ V}$$

$$V_1 = ?$$

$$V_1 = V_s - V_2$$

$$V_1 = 12 - 9.0$$

$$\underline{V_1 = 3.0 \text{ V}}$$

In the exam, this voltage is a mark of its own.

Worked example – then the resistance

A divider on a 12 V supply has $R_1 = 2.2 \text{ k}\Omega$ and an LDR as R_2 .
 $V_2 = 9.0 \text{ V}$. Calculate the resistance of the LDR.

$$V_1 = 3.0 \text{ V}$$

$$V_2 = 9.0 \text{ V}$$

$$R_1 = 2200 \text{ }\Omega$$

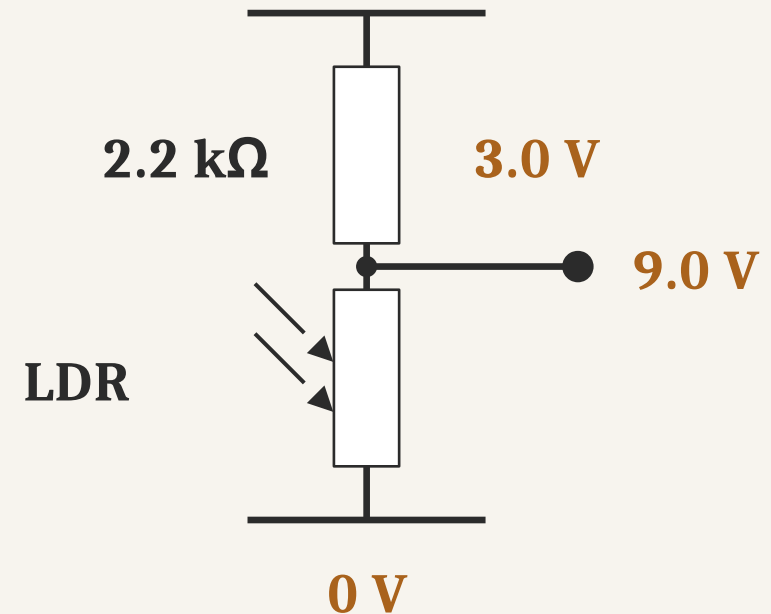
$$R_2 = ?$$

$$V_1/V_2 = R_1/R_2$$

$$3.0/9.0 = 2200/R_2$$

$$R_2 = 9.0 \times 2200/3.0$$

$$\underline{R_2 = 6600 \text{ }\Omega}$$



Show me — find the sensor's resistance

SHOW ME BOARDS

- 1 $V_s = 6 \text{ V}$, $R_1 = 1 \text{ k}\Omega$. An LDR is R_2 and $V_2 = 4 \text{ V}$. Find R_2 .
- 2 $V_s = 9 \text{ V}$, $R_2 = 3 \text{ k}\Omega$. A thermistor is R_1 and $V_2 = 3 \text{ V}$. Find R_1 .
- 3 $V_s = 5 \text{ V}$, $R_1 = 20 \text{ k}\Omega$. A thermistor is R_2 and $V_2 = 1 \text{ V}$. Find R_2 .
- 4 $V_s = 12 \text{ V}$ and $V_2 = 8 \text{ V}$. What is V_1 ?
- 5 $V_s = 10 \text{ V}$ and $V_2 = 2.5 \text{ V}$. What goes into the relationship as V_1 ?

Show me — find the sensor's resistance

SHOW ME BOARDS

EXTRA PRACTICE

- 1 $V_s = 12\text{ V}$, $R_1 = 4\text{ k}\Omega$. An LDR is R_2 and $V_2 = 8\text{ V}$. Find R_2 .
- 2 $V_s = 5\text{ V}$, $R_2 = 2\text{ k}\Omega$. An LDR is R_1 and $V_2 = 4\text{ V}$. Find R_1 .
- 3 $V_s = 6\text{ V}$, $R_1 = 2.2\text{ k}\Omega$. A thermistor is R_2 and $V_2 = 3\text{ V}$. Find R_2 .
- 4 $V_s = 9\text{ V}$ and $V_1 = 6.5\text{ V}$. What is V_2 ?
- 5 Your answer is $3.3333\text{ k}\Omega$. Write it to 2 significant figures.

Your turn – the practice questions

Booklet p.25-26 – PRACTICE: Voltage Dividers and Sensors

Answer questions 1 to 10 in your booklet.
The marks are printed beside each question.
Questions 4, 5 and 10 want a description –
resistance first, then V_2 .

Booklet p.25-26 – PRACTICE, Q1-10

PRACTICE — Answers, Q1-3

ANSWERS

Booklet p.25-26 — Voltage Dividers and Sensors

Q	The answer	Marks
1	a signal that can take any value in a range, and changes smoothly	1
2	to produce a smaller voltage from the supply voltage	1
3	$V_2 = (3 \times 440) \div 220 = 6 \text{ V}$	3

PRACTICE – Answers, Q4-6

ANSWERS

Booklet p.25-26 – Voltage Dividers and Sensors

Q	The answer	Marks
4	the LDR's resistance increases, so it takes more of the supply and V_2 decreases	2
5	the thermistor's resistance decreases, so V_2 across it decreases	2
6	$V_2 = (2 \times 3000) \div 2000 = 3 \text{ V}$	3

PRACTICE — Answers, Q7-8

ANSWERS

Booklet p.25-26 — Voltage Dividers and Sensors

Q	The answer	Marks
7	the two resistors share the supply, so some is always across R_1 — V_2 must be less	2
8	$V_1 = (8 \times 1000) \div 4000 = 2 \text{ V}$	3

PRACTICE — Answers, Q9-10

ANSWERS

Booklet p.25-26 — Voltage Dividers and Sensors

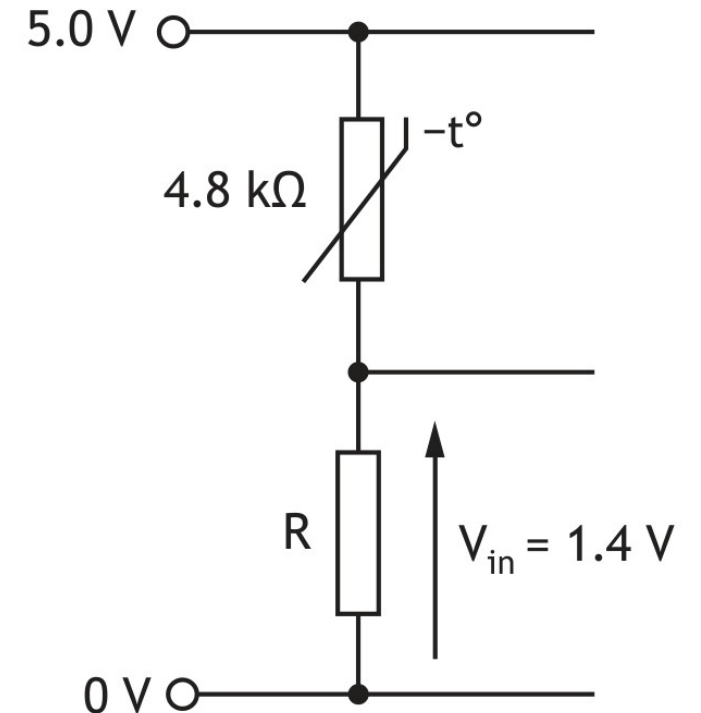
Q	The answer	Marks
9	an adjustable input signal — a volume control, a dimmer, or setting a trigger point	2
10	in the lower position, as R_2 , so V_2 rises as the light falls	1

Exam practice

National 5 Engineering Science 2024 Q10 – a kettle sensing circuit

Part	What it asks	Marks
10(b)	the name of this type of series circuit	1
10(c)	the resistance R in the circuit	4

Four marks: missing voltage, substitution, rearranging, answer.



LOOKING AHEAD

Transistors and Relays

Turning a changing voltage into a switch.

Plenary

- 1 What is a voltage divider made from?
- 2 Which resistor is V_2 taken across?
- 3 $R_1 = 2 \text{ k}\Omega$, $R_2 = 4 \text{ k}\Omega$, $V_1 = 5 \text{ V}$. Find V_2 .
- 4 An LDR is R_1 . Describe what happens to V_2 as it gets darker.
- 5 Why can V_2 never be bigger than the supply voltage?

- 1 What is an analogue signal?
- 2 Which resistor has the larger voltage across it?
- 3 $R_1 = 5 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $V_1 = 10 \text{ V}$. Find V_2 .
- 4 A thermistor is R_2 . Describe what happens to V_2 as it gets hotter.
- 5 $V_s = 6 \text{ V}$ and $V_2 = 2 \text{ V}$. What is V_1 ?

Extension

- 1 A divider has $100\ \Omega$ at the top and $100\ \text{k}\Omega$ at the bottom. Is V_2 close to the supply or close to 0 V? Explain.
- 2 A frost alarm must sound when it gets cold. Where does the thermistor go, and why not the other way?
- 3 Why is a voltage divider a better sensor circuit than measuring the sensor's resistance directly?

- 1 A divider has two $1\text{ M}\Omega$ resistors. What is V_2 as a fraction of the supply, and does the size of the resistors change that?
- 2 A freezer alarm must sound if the freezer warms up. Where does the thermistor go, and why?
- 3 A divider on a 5 V supply reads $V_2 = 5\text{ V}$. Suggest what is wrong with the circuit.

Course specification

N5 Engineering Science course specification v2.0, p.5

Analogue electronic control systems

- calculations of voltage, current, and unknown values in a fixed-voltage divider
- designing a voltage divider to provide an input signal for a control circuit
- interpreting information given of characteristics for an LDR and an NTC thermistor

Current in a divider uses Ohm's Law, from Lesson 2.